Features and Functions

Smart EON Multifunction Power Meter

Salient Features

- 128 samples / cycle
- 7 days data storage
- 10 secs battery backup to capture Power interuption event
- Multiple communication protocol options: 4G / RS 485 / LAN
- Remote configuration
- Auto Phase sequence adjustment

Functions

Measurements

- V, I, kW, kVAR, kVA, kWh, kVArh, kVAh, PF, Hz, V & I phase angle, Temperature, Residual Current
- 3Ph V & I, Unbalance V & I
- Latest 10 years, 12 months, 31 days historical energy

Communication / Data Sync Option

- Hardware communication: Standard 1 RS485 and 1 infrared communication
- Communication protocol: 4G / Modbus-RTU and Modbus-TCP
- Threshold setpoint for over-Voltage, under-Voltage, over-Current, under-Current, Phase loss, Max kW, Max temperature, leakage Current and Over load Kw

Power Quality Analysis

- Three Phase Voltage & Current
- 63rd Harmonic Analysis
- Voltage & Current Unbalance

Technical Specification

Power Supply	AC 85 to 265V, DC 100 to 300V	
Rated Current	5A (Standard CT)	
Rated Voltage	380V, 50Hz	
Digital Input	Wet contact, External power supply: 220V, open when <60V; closed when >140V Max. input: 300V	
Connection mode	3 phase 4 wire, 3 phase 3 wire	
Communication	4G/RS 485/LAN	
Residual Current	0 - 8000mA	
Temperature	0 - 150°C	
RS485	4800 to 38400bps	
Accuracy		
Voltage	Phase Voltage: 10V to 400V Line Voltage: 10V to 500V	0.2%
Primary Voltage	Max. 1000kV	-
Current	5mA to 6.5A	0.2%
Primary Current	Max. 1,00,000A	-
Frequency	40 to 70 Hz	0.1%
Active Energy	0 to 99999999.9 kWh	0.5%
Reactive Energy	0 to 99999999.9 kWh	2%
Active Power	Single Phase: 0 to ± 9999MW/Mvar	0.5%
Reactive Power	Total: 0 to ± 9999MW/Mvar	1.0%
Harmonic ratio	0% to 100%	B Level
Environment		
Operating Temperature	-25°C to +70°C	
Storage Temperature	-40°C~+85°C	
Humidity	5%~95% Non-condensing	
Electromagnetic compatibility		
Electrostatic discharge immunity test	IEC 61000-4-2:2001	
Radio Frequency electromagnetic field immunity test	IEC61000-4-3:2002	
Electrical fast transient immunity test	IEC 61000-4-4:2006	
Surge immunity test	IEC 61000-4-5 2005	
Injected Current immunity test	IEC 61000-4-6:2006	
Electromagnetic emission limit	CISPR22: 2006	